Problem 3.

Consider a <u>subspace W</u> of $\mathbb{R}^n$.

(a) Give a precise definition of the orthogonal complement, $W^\perp$ of W.

Sol-n:

Let W be a subspace of $\mathbb{R}^n$.

<u>Definition</u> ($W^\perp$, orthogonal complement of W):

The orthogonal complement, $W^\perp$ of a subspace W is

(1) $$S^\perp = \{ x \in \mathbb{R}^n : \langle x, y \rangle = 0 \ , \forall y \in W \}.$$

(b) Prove that $W^\perp$ is a subspace of $\mathbb{R}^n$. $\boxtimes$

Based on 3 properties of a subspace, let me verify $W^\perp$ meets them:

(i) $0 \in W^\perp$ because

$x \in W^\perp$, $\langle x, 0 \rangle = 0$, $0 \in W$ since it's a subspace.

(ii) Let $x_1, x_2 \in W^\perp$. Then by its defined way in (1)

(2) $\langle x_1, y \rangle = 0 \quad \forall y \in W$ &

(3) $\langle x_2 \ y \rangle = 0 \quad \forall y \in W$.

Now consider

$$\langle x_1 + x_2, y \rangle = \underbrace{\langle x_1, y \rangle}_{=0 \text{ by } (2)} + \underbrace{\langle x_2, y \rangle}_{=0 \text{ by } (3)} = 0 + 0 = 0 \Rightarrow$$

(4) $x_1 + x_2 \in W^\perp$.

(iii) Finally, let $x \in W^{\perp}$. $\Rightarrow$

(5) $\langle x, y \rangle = 0$, $\forall y \in W$.

And consider $c \in \mathbb{R}$ with

$\langle cx, y \rangle = c\langle x, y \rangle = c \cdot 0 = 0$, $\Rightarrow$

(6) $cx \in W^{\perp}$,

thus by (i), (ii), (iii), $W^{\perp}$ is a subspace of $\mathbb{R}^n$. $\boxtimes$

(c) Find an orthonormal basis of W,

$$W = \text{span}\{x, y\} \text{ wher } x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \ \& \ y = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

Sol-n. By Gram-Schmidt process, let me try to find the orthonormal basis $\{u_1, u_2\}$ s.t.

(7) $$W = \text{span}\{x, y\} = \text{span}\{u_1, u_2\}.$$

$$v_1 = x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \ \&$$

$$v_2 = y - \frac{\langle y, v_1 \rangle}{\langle v_1, v_1 \rangle} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} - \frac{\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} - \frac{2}{2} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

So we have orthogonal set $\{v_1, v_2\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$. Let's normalize it

(8) $$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \ \& \ u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Thus for (7), we have

$$W = \text{span}\{x, y\} = \text{span}\{u_1, u_2\} = \text{span}\left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}. \ \boxtimes$$

(d) Find the orthogonal projection of

(9) $$z = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

onto the subspace W in (c).

Sol-n: By the Orthogonal Decomposition Theorem (ODT) below, we can find $\text{proj}_W(z)$.

Theorem (ODT):
Let $W \subseteq \mathbb{R}^n$ be a subspace. Then, $\forall x \in \mathbb{R}^n$, $\exists$ unique $w \in W$, $\hat{w} \in W^\perp$ such that $x = w + \hat{w}$.
Moreover,

(10) $$w = \sum_{i=1}^{k} \langle x, u_i \rangle u_i$$

where $\{u_1, u_2, .. u_k\}$ is an orthonormal basis for W, & $\hat{w} = x - w$,
(w = "orthogonal projection of x onto W, i.e. $w = \text{proj}_W(x)$.).

Then by this Thm (ODT) & our obtained orthonormal basis in (8),

$$w = \text{proj}_W(z) = \sum_{i=1}^{2} \langle z, u_i \rangle \cdot u_i = \langle z, u_1 \rangle u_1 + \langle z, u_2 \rangle u_2 =$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \sqrt{2} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

(11) $$w = \text{proj}_W(z) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

$\boxtimes$